Scout: model routing evidence

Three local research reports behind the decision not to route Scout chat turns between Sonnet and Haiku. They were produced on Salman's machine and are not in the rag or swo-3.0 repositories.

Key figures

	Sonnet 5 (control)	Haiku 4.5 (arm)
Cost, same 288 turns	$4.10	$2.51 (–39%)
First visible text, p50	3.2 s	2.0 s
Catalogue asks that work well (gate ≥ 90%)	25/26 ✅	21/26 ❌
Same ask, consistent behaviour (gate)	✅	❌
Real teacher asks that work well	98/102	92/102
Reliability runs that work well	42/42	38/42
Judge mean, real asks	4.87	4.50

From the architecture review (§1.1, §2(c)): each model has its own prompt cache, and a mid-turn Haiku step wrote 21,333 tokens of cache ($0.029) where a warm Sonnet call costs about $0.0064. A router call adds about $0.002 and 1–2 s per turn to save about $0.002 of cache reads on a warm turn. Scout's deterministic `direct_dispatch` already handles fully specified asks with 0 misroutes over 112 eval and 58 real asks.

Not measured: Bedrock Intelligent Prompt Routing itself, or whether it supports Sonnet 5 and Haiku 4.5 with tool use and streaming.

Contents

1. Haiku 4.5 arm (all 288 turns on Haiku), 2026-09-15
2. Sonnet control (same 288 turns, same snapshot), 2026-09-15
3. Scout prompt architecture review, 2026-10-03

Teacher eval — `haiku-arm-92ae3a0`

- Target: Counter({'v2': 288}) · base ['http://127.0.0.1:8037'] · snapshot ['92ae3a0']
- Window (UTC): 2026-09-15T12:08:20+00:00 → 2026-09-15T12:16:43+00:00 · turns 288 · models {'us.anthropic.claude-haiku-4-5-20251001-v1:0': 288}
- System cost $2.505 (service-reported or estimated); judge rows 264
- Scorer notes: v2 question counts read `artifact.end.data.question_set` (targets.py, 2026-09-15, B5's emitted shape; before that fix a v2 run would have counted 0). checks.py is unchanged, so v1 counts are unaffected (pinned in check_targets.py on recorded v1 rows).
- Fixture-limited (excluded from gate denominators, kept in tables): 6 runs of ['cat.AS9.plan', 'cat.LP8.plan'] — the __probe article card has no page_content.source or grade; the same asks run on real cards in the `adapt_real_card` family.
- Harness: rag 3ba04cc (runner, checks, judge and report ran from a git archive of experiments/teacher_eval at this sha).
- Fast re-run (fast.sh): v2 at http://127.0.0.1:8037 (snapshot 92ae3a0), n=3, tasks: cat.LP5.plan,cat.LP8.plan,cat.LP1.plan,cat.LP1.agent,cat.LP3.plan,cat.LP3.agent,cat.CM2.plan,cat.CM4.plan,cat.CM5.plan,cat.CM9.plan,cat.AS9.plan,cat.AS8.plan,cat.AS2.plan,cat.. Gate rows cover only these tasks; read the per-task diff.
- Baseline v2-full-4201d6f is shown with its frozen judge scores; corrected baselines (re-judge overlays) are eval/v2-full-4201d6f.corrected.md and eval/baseline-v1.corrected.md.

Gates

Gate (ScoutCoreV2 §1)	Measured	Target	Met ?	How
Catalogue asks that work well	21/26 (81%)	≥ 90%	❌	ask works well = majority of runs in every mode pass checks and judge overall ≥ 4 with did_the_task
Lesson plan classroom-ready	23/24 (96%)	≥ 95%	✅	all lesson-plan runs: CTO ask + R2 variants (both modes) + R5 LP on 8 articles; works well = checks pass and judge overall ≥ 4 with did_the_task
Requested question count delivered	9/9 (100%)	100%	✅	exact count, structured questions or numbered items
Article Q&A damaged after generation	0/24 measurable (0%) of 24 Q&A turns	0%	✅	final reply < 80% of the model's raw text
Distinct behaviours for the same ask (n=10)	max 1; 13/14 asks ≤ 2 and all acceptable	≤ 2, all acceptable	❌	reliability family, behaviour signature over outcome/tools/action/args/artifact kinds
Time to first visible text, p50	all 2.0s (warm 1.9s, cold 2.1s; n=253)	< 2 s	✅	ttft_any: first text.delta / artifact.delta (v1: whole reply); cold = first use of a card in the run
(info) First prose token, p50 / p95	2.0s / 3.2s (n=253; turns with no prose at all: 29)	—	—	ttft_prose: first text.delta only; not a gate
Any turn failing on a timeout	0/282	never	✅	error mentions timeout, outcome idle_timeout, or ≥180 s

Families

Family	Tasks	Runs	Det. pass	Works well	Judge mean (n)	Behaviours/task (mean, max)	TTFT any p50/p95	TTFT prose p50/p95	Total p50/p95	Cost
catalogue	30	90	84/90 (93%)	70/90 (78%)	4.06 (90)	1.3, 3	2.0s/3.5s	2.0s/3.5s	10.0s/30.6s	$0.913
lesson_plan	6	18	18/18 (100%)	17/18 (94%)	3.94 (18)	1.2, 2	1.9s/2.1s	1.9s/2.1s	25.8s/41.1s	$0.445
r5	10	30	30/30 (100%)	30/30 (100%)	4.80 (30)	1.0, 1	1.9s/2.6s	1.9s/2.6s	7.2s/35.8s	$0.479
reliability	14	42	39/42 (93%)	38/42 (90%)	4.79 (24)	1.0, 1	2.0s/3.2s	2.0s/3.2s	6.1s/10.9s	$0.212
real_asks	34	102	102/102 (100%)	92/102 (90%)	4.50 (102)	1.0, 2	2.0s/2.2s	2.0s/2.2s	5.6s/8.7s	$0.456

Diff vs `v2-full-4201d6f`

Task	before well	after well	before judge	after judge
arc.AD1.g4	3/3	—	—	—
arc.AD1.g45	3/3	—	—	—
arc.AD2.g4	3/3	—	—	—
arc.AD2.g45	3/3	—	—	—
arc.AD3.g4	3/3	—	—	—
arc.AD3.g45	3/3	—	—	—
arc.AD4.g4	3/3	—	—	—
arc.AD4.g45	3/3	—	—	—
arc.AD5.g4	3/3	—	—	—
arc.AD5.g45	3/3	—	—	—
cat.AD1.agent	3/3	—	—	—
cat.AD1.plan	3/3	—	—	—
cat.AD2.agent	3/3	—	—	—
cat.AD2.plan	3/3	—	—	—
cat.AD3.agent	3/3	—	—	—
cat.AD3.plan	3/3	—	—	—
cat.AD4.agent	3/3	—	—	—
cat.AD4.plan	3/3	—	—	—
cat.AD5.agent	3/3	—	—	—
cat.AD5.plan	3/3	—	—	—
cat.APP1.plan	0/3	0/3	1.7	2.3
cat.APP2.agent	3/3	—	—	—
cat.APP3.agent	3/3	—	—	—
cat.APP4.plan	0/3	3/3	2.0	4.0
cat.APP5.plan	0/3	0/3	2.3	2.0
cat.APP6.plan	0/3	—	3.0	—
cat.APP7.agent	3/3	—	—	—
cat.AS1.plan	2/3	—	4.3	—
cat.AS2.plan	3/3	3/3	4.7	5.0
cat.AS3.plan	3/3	—	4.7	—
cat.AS4.plan	0/3	3/3	2.7	5.0
cat.AS5.plan	3/3	—	5.0	—
cat.AS6.plan	3/3	—	5.0	—
cat.AS7.plan	1/3	—	3.3	—
cat.AS8.plan	3/3	1/3	4.0	3.0

Task	before well	after well	before judge	after judge
cat.AS9.plan	0/3	0/3	2.3	2.0
cat.CD1.plan	3/3	3/3	4.3	4.3
cat.CD2.plan	3/3	—	5.0	—
cat.CD3.plan	3/3	—	4.3	—
cat.CD4.plan	3/3	—	4.0	—
cat.CD5.plan	3/3	—	4.7	—
cat.CD6.plan	3/3	1/3	4.0	3.3
cat.CD7.plan	1/3	3/3	3.3	4.0
cat.CD8.plan	3/3	3/3	4.0	4.0
cat.CM1.plan	3/3	—	4.7	—
cat.CM10.plan	3/3	—	5.0	—
cat.CM2.plan	3/3	3/3	4.0	4.7
cat.CM3.plan	0/3	—	3.3	—
cat.CM4.plan	3/3	3/3	4.7	4.7
cat.CM5.plan	3/3	3/3	4.0	4.3
cat.CM6.plan	3/3	—	5.0	—
cat.CM7.plan	3/3	—	5.0	—
cat.CM8.plan	3/3	—	5.0	—
cat.CM9.plan	3/3	3/3	4.7	4.7
cat.CO1.plan	3/3	3/3	4.3	4.7
cat.CO2.plan	3/3	2/3	4.0	4.3
cat.CO3.plan	1/3	—	2.3	—
cat.CO4.plan	1/3	3/3	3.3	4.7
cat.LP1.agent	1/3	3/3	4.3	4.3
cat.LP1.plan	0/3	3/3	4.7	4.3
cat.LP10.plan	3/3	—	5.0	—
cat.LP11.plan	3/3	—	4.0	—
cat.LP2.agent	0/3	—	4.7	—
cat.LP2.plan	0/3	—	4.0	—
cat.LP3.agent	0/3	2/3	4.0	4.3
cat.LP3.plan	0/3	3/3	4.0	4.3
cat.LP4.agent	0/3	—	4.0	—
cat.LP4.plan	0/3	—	4.3	—
cat.LP5.plan	3/3	3/3	4.0	4.7
cat.LP6.plan	3/3	—	4.7	—
cat.LP7.plan	3/3	—	4.0	—
cat.LP8.plan	0/3	1/3	2.7	2.7
cat.LP9.plan	3/3	—	4.3	—
cat.OE1.plan	3/3	3/3	5.0	5.0
cat.OE2.plan	3/3	3/3	4.7	4.7
cat.OE3.plan	3/3	3/3	5.0	4.7
cat.OE4.plan	3/3	—	5.0	—
cat.OE5.plan	0/3	—	—	—
cat.OE6.plan	0/3	—	—	—
cat.ST1.plan	0/3	3/3	2.3	4.0
cat.ST2.plan	3/3	3/3	4.3	4.7
cat.ST3.plan	1/3	0/3	3.3	3.0
lp.CTO.agent	0/3	3/3	4.7	4.0
lp.CTO.plan	0/3	3/3	4.0	4.0
lp.LP1b.agent	0/3	3/3	4.0	4.3
lp.LP1b.plan	0/3	3/3	4.0	4.0
lp.LPR1.agent	0/3	3/3	3.0	4.0
lp.LPR1.plan	0/3	2/3	2.7	3.3
r5.122633.LP	0/3	3/3	4.7	5.0
r5.122633.LV7to5	3/3	—	4.0	—
r5.122633.MAIN	3/3	—	5.0	—
r5.122633.QG	3/3	—	5.0	—
r5.122633.VOCAB	3/3	3/3	4.7	5.0
r5.122633.WHY	3/3	—	5.0	—
r5.29060.LP	0/3	—	4.0	—
r5.29060.LV4to2	3/3	—	4.0	—
r5.29060.LV4to6	3/3	—	4.7	—
r5.29060.MAIN	3/3	3/3	5.0	4.0
r5.29060.QG	3/3	—	4.7	—
r5.29060.VOCAB	3/3	—	4.3	—
r5.29060.WHY	3/3	3/3	5.0	5.0
r5.29189.LP	1/3	—	4.7	—
r5.29189.LV3to1	3/3	—	5.0	—
r5.29189.MAIN	3/3	—	5.0	—
r5.29189.QG	3/3	3/3	4.7	5.0
r5.29189.VOCAB	3/3	—	5.0	—
r5.29189.WHY	3/3	—	5.0	—
r5.41150.LP	1/3	—	4.7	—
r5.41150.LV1to3	3/3	—	4.0	—
r5.41150.MAIN	3/3	—	5.0	—
r5.41150.QG	3/3	—	5.0	—
r5.41150.VOCAB	3/3	3/3	4.3	5.0
r5.41150.WHY	3/3	—	5.0	—
r5.45741.LP	0/3	—	4.0	—

Task	before well	after well	before judge	after judge
r5.45741.LV5to3	3/3	—	4.0	—
r5.45741.MAIN	3/3	—	4.3	—
r5.45741.QG	3/3	—	4.7	—
r5.45741.VOCAB	3/3	—	4.7	—
r5.45741.WHY	2/3	3/3	3.7	4.7
r5.46270.LP	0/3	3/3	4.3	4.3
r5.46270.LV6to4	3/3	—	4.0	—
r5.46270.LV6to8	3/3	—	5.0	—
r5.46270.MAIN	3/3	—	5.0	—
r5.46270.QG	3/3	—	4.3	—
r5.46270.VOCAB	3/3	—	4.7	—
r5.46270.WHY	3/3	—	4.0	—
r5.92658.LP	1/3	—	4.3	—
r5.92658.LV2to4	2/3	—	3.7	—
r5.92658.MAIN	3/3	3/3	4.7	5.0
r5.92658.QG	3/3	—	5.0	—
r5.92658.VOCAB	3/3	—	4.3	—
r5.92658.WHY	3/3	—	4.7	—
r5.94818.LP	0/3	—	4.7	—
r5.94818.LV8to10	3/3	—	4.7	—
r5.94818.LV8to6	3/3	—	4.7	—
r5.94818.MAIN	3/3	—	4.3	—
r5.94818.QG	3/3	3/3	4.7	5.0
r5.94818.VOCAB	3/3	—	5.0	—
r5.94818.WHY	3/3	—	5.0	—
ra.RA02.agent	—	3/3	—	5.0
ra.RA03.agent	—	3/3	—	5.0
ra.RA04.agent	—	3/3	—	5.0
ra.RA05.agent	—	1/3	—	3.0
ra.RA07.agent	—	3/3	—	4.7
ra.RA09.agent	—	3/3	—	4.7
ra.RA11.agent	—	3/3	—	3.7
ra.RA13.agent	—	2/3	—	3.7
ra.RA14.agent	—	1/3	—	3.3
ra.RA15.agent	—	3/3	—	4.3
ra.RA16.agent	—	1/3	—	3.7
ra.RA17.agent	—	3/3	—	5.0
ra.RA18.agent	—	3/3	—	5.0
ra.RA20.agent	—	3/3	—	5.0
ra.RA21.agent	—	3/3	—	4.0
ra.RA23.agent	—	3/3	—	5.0
ra.RA24.agent	—	3/3	—	5.0
ra.RA25.agent	—	3/3	—	4.7
ra.RA26.agent	—	3/3	—	5.0
ra.RA27.agent	—	3/3	—	5.0
ra.RA28.agent	—	3/3	—	4.3
ra.RA29.agent	—	3/3	—	4.3
ra.RA30.agent	—	3/3	—	5.0
ra.RA31.agent	—	3/3	—	5.0
ra.RA32.agent	—	3/3	—	5.0
ra.RA33.agent	—	3/3	—	5.0
ra.RA35.agent	—	3/3	—	4.3
ra.RA36.agent	—	3/3	—	5.0
ra.RA37.agent	—	3/3	—	4.3
ra.RA38.agent	—	0/3	—	1.0
ra.RA39.agent	—	3/3	—	5.0
ra.RA40.agent	—	3/3	—	5.0
ra.RA41.agent	—	3/3	—	5.0
ra.RA42.agent	—	3/3	—	5.0
rel.A01	10/10	3/3	4.8	5.0
rel.A02	2/10	—	4.4	—
rel.A03	10/10	3/3	5.0	5.0
rel.A04	10/10	—	—	—
rel.A05	10/10	3/3	—	—
rel.A06	10/10	—	—	—
rel.C01	10/10	3/3	—	—
rel.G01	10/10	3/3	5.0	5.0
rel.G02	10/10	3/3	5.0	5.0
rel.G03	10/10	—	—	—
rel.N01	10/10	3/3	5.0	5.0
rel.P01	10/10	3/3	5.0	5.0
rel.P02	10/10	3/3	—	—
rel.S01	0/10	2/3	2.9	3.7
rel.S02	10/10	3/3	—	—
rel.T01	10/10	0/3	—	—
rel.T02	10/10	3/3	5.0	4.7
rel.X01	10/10	3/3	—	—

Turn outcomes (v2)

Family	complete	paused	stopped	error	idle_timeout	tool_error	no_turn_end
catalogue	89	1	0	0	0	0	0
lesson_plan	18	0	0	0	0	0	0
r5	30	0	0	0	0	0	0
real_asks	69	33	0	0	0	0	0
reliability	36	12	0	0	0	0	0

First content (v2 telemetry `first_content_ms`; not a gate)

Turns with a value: 259; turns whose telemetry row lacks the key: 0.

Measure	p50	p95	n
first_content_ms (telemetry, all)	2.0s	3.2s	259
first_content_ms (cold / warm p50)	2.1s / 1.9s	—	44 / 215
ttft_any_ms (eval, client-side, same turns)	2.0s	3.2s	259

Engine stage timings (v2, from artifact.end, seconds from tool start; not a gate)

Engine	n	resources p50/p95	engine_start p50/p95	first_token p50/p95	first_item p50/p95	draft p50/p95	write p50/p95	check p50/p95	verify p50/p95	total p50/p95	cache_read (median sum)
generate_questions	12	—	0.0/0.0	1.9/2.1	3.4/3.7	—	7.5/8.4	—	2.0/2.3	10.6/16.2	0.0
write_artifact (document)	2	0.0/0.0	0.0/0.0	1.8/1.8	—	9.2/13.5	—	2.3/2.4	—	11.6/15.8	1188.0
write_artifact (lesson_plan)	18	0.0/0.0	0.0/0.0	1.9/2.7	—	19.7/25.3	—	1.1/7.4	—	22.5/32.5	1098.0
write_artifact (weekly_plan)	5	0.0/0.8	0.0/0.8	1.8/2.8	—	26.0/27.1	—	4.0/12.6	—	29.8/38.8	1716

Context degradation (v2, silent to the teacher)

Telemetry rows joined: 288/288 v2 turns (missing = no tokenz row for that turn_id).

Rate	Measured	How
Resource lookup failed	0/288 (0.0%)	`ctx_resource_lookup_failed` truthy, over turns that report the field
History summary failed	0/288 (0.0%)	`ctx_summary` equals failed, over turns that report the field
Article truncated (notice shown)	0/117 (0.0%)	`ctx_article_notices > 0`, over turns with article text

Tool-name corrections (v2, the model called a capability id as a tool name)

Key `tool_name_corrections` (names `tool_names_corrected`): 0/288 turns with ≥1 correction (0.0%); 0 corrections in total. Unjoined turns (no telemetry row): 0. **Contract breaches (row present, key missing or non-integer): 0** ✅.

Family	Turns	With correction	Corrections
catalogue	90	0 (0%)	0
lesson_plan	18	0 (0%)	0
r5	30	0 (0%)	0
real_asks	102	0 (0%)	0
reliability	48	0 (0%)	0

Outcome	Turns	With correction
complete	242	0 (0%)
paused	46	0 (0%)

LP8: old vs corrected criterion

Corrected 2026-09-15 (coordinator-approved): a card with no current week cannot resolve "plan next week", so asking which week with a pending_prompt whose options name a real week is correct; a guessed-week plan, a hedge or a prose question fails.

On an article card that names the next week (nextWeekId), the pass is a weekly plan for that week naming at least 2 of its real titles; asking with real options there is the soft column, not a pass.

Task	Turns	Old criterion pass	Corrected pass	Asked, real options	Asked, no real option	Planned a guessed week	Planned the named next week	Asked, real options (soft)	Hedge / other
cat.LP8.plan	3	2	1	1	0	2	0	0	0

Per task

Task	Ask	n	Det.	Well	Judge	Behaviours	Common failures	Example (first failing run)
cat.APP1.plan	how do I assign this article to my class?	3	2	0	2.3	1	no_unsupported_app_claim×1	complete: That's done from a different page. Go back to your classroom's main area (or the Publicati
cat.APP4.plan	how do I create a new classroom?	3	3	3	4.0	1		
cat.APP5.plan	how do I add a student to my class?	3	3	0	2.0	1		complete: On the Studies Weekly app, you add a student through the Classroom Settings page, not from
cat.AS2.plan	make a quiz on this article	3	3	3	5.0	1		
cat.AS4.plan	make question 3 easier	3	3	3	5.0	1		
cat.AS8.plan	write feedback comments for the students who sco	3	2	1	3.0	3	facts_all×1	complete: I'll pull up the resources for this assessment so I can write meaningful feedback tied to
cat.AS9.plan	create an assessment covering week 1	3	2	0	2.0	2	questions_present×1	complete: I'll create an assessment covering Week 1: The First Peoples. I'll create a comprehensive
cat.CD1.plan	who is falling behind?	3	3	3	4.3	1		
cat.CD6.plan	based on these results, what should I reteach?	3	3	1	3.3	1		complete: Based on your class results for Week 5 Assessment, here's what the data suggests: **The c
cat.CD7.plan	how is Student 1 doing?	3	3	3	4.0	1		
cat.CD8.plan	what do I need to grade?	3	3	3	4.0	1		
cat.CM2.plan	create a vocabulary activity from this article	3	3	3	4.7	1		
cat.CM4.plan	make a graphic organizer for this article	3	3	3	4.7	1		

Task	Ask	n	Det.	Well	Judge	Behaviours	Common failures	Example (first failing run)
cat.CM5.plan	create a rubric for a paragraph response about t	3	3	3	4.3	1		
cat.CM9.plan	make a vocabulary list with kid-friendly definit	3	3	3	4.7	1		
cat.CO1.plan	write an email to parents about what we're learn	3	3	3	4.7	1		
cat.CO2.plan	draft a note to Student 8 reminding them to turn	3	3	2	4.3	1		complete: I'll write a note to Student 8 reminding them to complete and submit the assessment. Your
cat.CO4.plan	write a short newsletter paragraph about what ou	3	3	3	4.7	1		
cat.LP1.agent	create a lesson plan for the article	3	3	3	4.3	1		
cat.LP1.plan	create a lesson plan for the article	3	3	3	4.3	1		
cat.LP3.agent	plan a lesson using this	3	2	2	4.3	2	no_unprompted_duration×1	complete: I'll write a lesson plan from this article about producers and consumers. No published le
cat.LP3.plan	plan a lesson using this	3	3	3	4.3	1		
cat.LP5.plan	How should I differentiate this article for my s	3	3	3	4.7	1		
cat.LP8.plan	Help me plan next week from this publication.	3	1	1	2.7	3	asks_which_week_with_real_options×2	complete: I'll look up what's in next week and then make you a week plan. Looking at the page, you'
cat.OE1.plan	what is a shortage?	3	3	3	5.0	1		
cat.OE2.plan	why do prices go up when there is a shortage?	3	3	3	4.7	1		
cat.OE3.plan	give me a real-world example of producers and co	3	3	3	4.7	1		
cat.ST1.plan	what standards does this article cover?	3	3	3	4.0	1		
cat.ST2.plan	what standards are covered this week?	3	3	3	4.7	1		
cat.ST3.plan	align this article to Common Core standards	3	3	0	3.0	2		complete: I'll look up the Common Core standards that align with this article's content about produc
lp.CTO.agent	Create a lesson plan for this article.	3	3	3	4.0	1		
lp.CTO.plan	Create a lesson plan for this article.	3	3	3	4.0	1		
lp.LP1b.agent	create a lesson plan for this article	3	3	3	4.3	1		
lp.LP1b.plan	create a lesson plan for this article	3	3	3	4.0	1		
lp.LPR1.agent	create a lesson plan for this week	3	3	3	4.0	1		
lp.LPR1.plan	create a lesson plan for this week	3	3	2	3.3	2		complete: I'll write a lesson plan for Week 3 — Water on Earth. Looking at the Teacher Resources pa
r5.122633.LP	Create a lesson plan for this article.	3	3	3	5.0	1		
r5.122633.VOCAB	What does "empirical" mean in this article? Expl	3	3	3	5.0	1		
r5.29060.MAIN	What is the main idea of this article? Give me a	3	3	3	4.0	1		
r5.29060.WHY	Why are fossils helpful to paleontologists?	3	3	3	5.0	1		
r5.29189.QG	Create 5 questions for my class from this articl	3	3	3	5.0	1		
r5.41150.VOCAB	What does "tilt" mean in this article? Explain i	3	3	3	5.0	1		
r5.45741.WHY	Why were the Spanish and French fleets fighting	3	3	3	4.7	1		
r5.46270.LP	Create a lesson plan for this article.	3	3	3	4.3	1		
r5.92658.MAIN	What is the main idea of this article? Give me a	3	3	3	5.0	1		
r5.94818.QG	Create 5 questions for my class from this articl	3	3	3	5.0	1		
ra.RA02.agent	Show inactive classrooms	3	3	3	5.0	1		
ra.RA03.agent	search this page for Demo	3	3	3	5.0	1		
ra.RA04.agent	Show active classrooms	3	3	3	5.0	1		
ra.RA05.agent	can you fill it for me	3	3	1	3.0	1		complete: I'd be happy to help you finish setting up this classroom. To fill in what's missing, I ne
ra.RA07.agent	Are there any ties in the leaderboard, and who i	3	3	3	4.7	1		
ra.RA09.agent	is anything else still needed before I can see t	3	3	3	4.7	1		
ra.RA11.agent	what is in the Training category?	3	3	3	3.7	1		
ra.RA13.agent	How many separate pieces of customized content h	3	3	2	3.7	2		complete: I can see the Standards Overview for your Customized Content Report, which shows performan
ra.RA14.agent	What are my three most recent conversations with	3	3	1	3.3	1		complete: I don't have access to your conversation history or saved conversations—that information i
ra.RA15.agent	Has anybody left this class? If so tell me who,	3	3	3	4.3	1		
ra.RA16.agent	Which of my active classrooms has the most publi	3	3	1	3.7	1		complete: I can see your three active classrooms on this page, but the classroom list doesn't show h
ra.RA17.agent	Walk me through what happens on the fifth screen	3	3	3	5.0	1		
ra.RA18.agent	what is this draft called?	3	3	3	5.0	1		
ra.RA20.agent	what is in unit 2?	3	3	3	5.0	1		
ra.RA21.agent	Rank this pupil's weeks from strongest to weakes	3	3	3	4.0	1		
ra.RA23.agent	according to this page, why is the profile form	3	3	3	5.0	1		
ra.RA24.agent	Put the whole list in order by one property Grad	3	3	3	5.0	1		
ra.RA25.agent	Which fields on this page can I edit and which a	3	3	3	4.7	1		
ra.RA26.agent	create and assign this customized content now	3	3	3	5.0	1		
ra.RA27.agent	How much of the publication has the class finish	3	3	3	5.0	1		
ra.RA28.agent	list my highlights	3	3	3	4.3	1		
ra.RA29.agent	If I answer a quiz question while I am impersona	3	3	3	4.3	1		
ra.RA30.agent	How many things are in that list altogether?	3	3	3	5.0	1		
ra.RA31.agent	which assessment went worst?	3	3	3	5.0	1		

Task	Ask	n	Det.	Well	Judge	Behaviours	Common failures	Example (first failing run)
ra.RA32.agent	how many students have handed this in?	3	3	3	5.0	1		
ra.RA33.agent	pause the video	3	3	3	5.0	1		
ra.RA35.agent	Which kinds of content is this student answering	3	3	3	4.3	1		
ra.RA36.agent	open the previous school year notes	3	3	3	5.0	1		
ra.RA37.agent	Tell me about clouds in general, weather pattern	3	3	3	4.3	1		
ra.RA38.agent	How many students have completed this, how many	3	3	0	1.0	1		paused, action grade.submissionStatus:
ra.RA39.agent	switch to my other classroom	3	3	3	5.0	1		
ra.RA40.agent	open the first rubric you can find on this page	3	3	3	5.0	1		
ra.RA41.agent	list the teacher resource panels on this page	3	3	3	5.0	1		
ra.RA42.agent	hide week 1 assessments from students	3	3	3	5.0	1		
rel.A01	What is this article about?	3	3	3	5.0	1		
rel.A03	Create 3 multiple choice questions from this art	3	3	3	5.0	1		
rel.A05	Find me articles about the water cycle.	3	3	3	—	1		
rel.C01	Delete dsfsdf.	3	3	3	—	1		
rel.G01	What do I have waiting to be graded?	3	3	3	5.0	1		
rel.G02	What is the class average this week?	3	3	3	5.0	1		
rel.N01	how many submissions are waiting to be graded in	3	3	3	5.0	1		
rel.P01	how many students have a verified family email?	3	3	3	5.0	1		
rel.P02	drop Student 4 from this class	3	3	3	—	1		
rel.S01	How do I add a co-teacher to my classroom?	3	3	2	3.7	1		complete: Co-teachers are invited on the People page. I can open that page for you now. Want me to
rel.S02	Change the grade to Fifth Grade.	3	3	3	—	1		
rel.T01	take me to week 5	3	0	0	—	1	action_proposed×3	paused, action publication.goToWeek:
rel.T02	what is hidden from students right now?	3	3	3	4.7	1		
rel.X01	Find me articles about the branches of governmen	3	3	3	—	1		

Teacher eval — `haiku-ctl-92ae3a0`

- Target: Counter({'v2': 288}) · base ['http://127.0.0.1:8036'] · snapshot ['92ae3a0']
- Window (UTC): 2026-09-15T12:08:20+00:00 → 2026-09-15T12:18:08+00:00 · turns 288 · models {'us.anthropic.claude-sonnet-5': 288}
- System cost $4.104 (service-reported or estimated); judge rows 264
- Scorer notes: v2 question counts read `artifact.end.data.question_set` (targets.py, 2026-09-15, B5's emitted shape; before that fix a v2 run would have counted 0). checks.py is unchanged, so v1 counts are unaffected (pinned in check_targets.py on recorded v1 rows).
- Fixture-limited (excluded from gate denominators, kept in tables): 6 runs of ['cat.AS9.plan', 'cat.LP8.plan'] — the __probe article card has no page_content.source or grade; the same asks run on real cards in the `adapt_real_card` family.
- Harness: rag 3ba04cc (runner, checks, judge and report ran from a git archive of experiments/teacher_eval at this sha).
- Fast re-run (fast.sh): v2 at http://127.0.0.1:8036 (snapshot 92ae3a0), n=3, tasks: cat.LP5.plan,cat.LP8.plan,cat.LP1.plan,cat.LP1.agent,cat.LP3.plan,cat.LP3.agent,cat.CM2.plan,cat.CM4.plan,cat.CM5.plan,cat.CM9.plan,cat.AS9.plan,cat.AS8.plan,cat.AS2.plan,cat.. Gate rows cover only these tasks; read the per-task diff.
- Baseline v2-full-4201d6f is shown with its frozen judge scores; corrected baselines (re-judge overlays) are eval/v2-full-4201d6f.corrected.md and eval/baseline-v1.corrected.md.

Gates

Gate (ScoutCoreV2 §1)	Measured	Target	Met ?	How
Catalogue asks that work well	25/26 (96%)	≥ 90%	✅	ask works well = majority of runs in every mode pass checks and judge overall ≥ 4 with did_the_task
Lesson plan classroom-ready	23/24 (96%)	≥ 95%	✅	all lesson-plan runs: CTO ask + R2 variants (both modes) + R5 LP on 8 articles; works well = checks pass and judge overall ≥ 4 with did_the_task
Requested question count delivered	9/9 (100%)	100%	✅	exact count, structured questions or numbered items
Article Q&A damaged after generation	0/24 measurable (0%) of 24 Q&A turns	0%	✅	final reply < 80% of the model's raw text
Distinct behaviours for the same ask (n=10)	max 2; 14/14 asks ≤ 2 and all acceptable	≤ 2, all acceptable	✅	reliability family, behaviour signature over outcome/tools/action/args/artifact kinds
Time to first visible text, p50	all 3.2s (warm 3.2s, cold 3.4s; n=240)	< 2 s	❌	ttft_any: first text.delta / artifact.delta (v1: whole reply); cold = first use of a card in the run
(info) First prose token, p50 / p95	3.2s / 7.1s (n=240; turns with no prose at all: 42)	—	—	ttft_prose: first text.delta only; not a gate
Any turn failing on a timeout	0/282	never	✅	error mentions timeout, outcome idle_timeout, or ≥180 s

Families

Family	Tasks	Runs	Det. pass	Works well	Judge mean (n)	Behaviours/task (mean, max)	TTFT any p50/p95	TTFT prose p50/p95	Total p50/p95	Cost
catalogue	30	90	87/90 (97%)	80/90 (89%)	4.29 (90)	1.2, 3	3.8s/7.6s	3.8s/7.6s	13.3s/27.3s	$1.410
lesson_plan	6	18	18/18 (100%)	17/18 (94%)	4.28 (18)	1.3, 2	1.8s/6.7s	1.8s/6.7s	25.5s/41.5s	$0.480
r5	10	30	30/30 (100%)	30/30 (100%)	4.90 (30)	1.0, 1	3.0s/3.6s	3.0s/3.6s	10.0s/29.9s	$0.601
reliability	14	42	42/42 (100%)	42/42 (100%)	4.96 (24)	1.1, 2	3.3s/7.8s	3.3s/7.8s	7.3s/13.1s	$0.484
real_asks	34	102	100/102 (98%)	98/102 (96%)	4.87 (102)	1.1, 2	3.2s/5.6s	3.2s/5.6s	7.0s/12.7s	$1.129

Diff vs `v2-full-4201d6f`

Task	before well	after well	before judge	after judge
arc.AD1.g4	3/3	—	—	—
arc.AD1.g45	3/3	—	—	—
arc.AD2.g4	3/3	—	—	—
arc.AD2.g45	3/3	—	—	—
arc.AD3.g4	3/3	—	—	—
arc.AD3.g45	3/3	—	—	—
arc.AD4.g4	3/3	—	—	—
arc.AD4.g45	3/3	—	—	—
arc.AD5.g4	3/3	—	—	—
arc.AD5.g45	3/3	—	—	—
cat.AD1.agent	3/3	—	—	—
cat.AD1.plan	3/3	—	—	—
cat.AD2.agent	3/3	—	—	—
cat.AD2.plan	3/3	—	—	—
cat.AD3.agent	3/3	—	—	—
cat.AD3.plan	3/3	—	—	—
cat.AD4.agent	3/3	—	—	—
cat.AD4.plan	3/3	—	—	—
cat.AD5.agent	3/3	—	—	—
cat.AD5.plan	3/3	—	—	—
cat.APP1.plan	0/3	3/3	1.7	4.3
cat.APP2.agent	3/3	—	—	—
cat.APP3.agent	3/3	—	—	—
cat.APP4.plan	0/3	2/3	2.0	3.7
cat.APP5.plan	0/3	3/3	2.3	4.3
cat.APP6.plan	0/3	—	3.0	—
cat.APP7.agent	3/3	—	—	—
cat.AS1.plan	2/3	—	4.3	—
cat.AS2.plan	3/3	3/3	4.7	4.7
cat.AS3.plan	3/3	—	4.7	—
cat.AS4.plan	0/3	3/3	2.7	4.3
cat.AS5.plan	3/3	—	5.0	—
cat.AS6.plan	3/3	—	5.0	—
cat.AS7.plan	1/3	—	3.3	—
cat.AS8.plan	3/3	3/3	4.0	4.7

Task	before well	after well	before judge	after judge
cat.AS9.plan	0/3	0/3	2.3	2.3
cat.CD1.plan	3/3	3/3	4.3	4.7
cat.CD2.plan	3/3	—	5.0	—
cat.CD3.plan	3/3	—	4.3	—
cat.CD4.plan	3/3	—	4.0	—
cat.CD5.plan	3/3	—	4.7	—
cat.CD6.plan	3/3	3/3	4.0	5.0
cat.CD7.plan	1/3	2/3	3.3	3.7
cat.CD8.plan	3/3	3/3	4.0	4.3
cat.CM1.plan	3/3	—	4.7	—
cat.CM10.plan	3/3	—	5.0	—
cat.CM2.plan	3/3	3/3	4.0	4.0
cat.CM3.plan	0/3	—	3.3	—
cat.CM4.plan	3/3	3/3	4.7	5.0
cat.CM5.plan	3/3	3/3	4.0	4.3
cat.CM6.plan	3/3	—	5.0	—
cat.CM7.plan	3/3	—	5.0	—
cat.CM8.plan	3/3	—	5.0	—
cat.CM9.plan	3/3	3/3	4.7	4.7
cat.CO1.plan	3/3	3/3	4.3	4.7
cat.CO2.plan	3/3	3/3	4.0	4.0
cat.CO3.plan	1/3	—	2.3	—
cat.CO4.plan	1/3	3/3	3.3	4.7
cat.LP1.agent	1/3	3/3	4.3	4.0
cat.LP1.plan	0/3	3/3	4.7	4.0
cat.LP10.plan	3/3	—	5.0	—
cat.LP11.plan	3/3	—	4.0	—
cat.LP2.agent	0/3	—	4.7	—
cat.LP2.plan	0/3	—	4.0	—
cat.LP3.agent	0/3	3/3	4.0	4.0
cat.LP3.plan	0/3	2/3	4.0	4.0
cat.LP4.agent	0/3	—	4.0	—
cat.LP4.plan	0/3	—	4.3	—
cat.LP5.plan	3/3	3/3	4.0	5.0
cat.LP6.plan	3/3	—	4.7	—
cat.LP7.plan	3/3	—	4.0	—
cat.LP8.plan	0/3	1/3	2.7	3.3
cat.LP9.plan	3/3	—	4.3	—
cat.OE1.plan	3/3	3/3	5.0	5.0
cat.OE2.plan	3/3	3/3	4.7	4.7
cat.OE3.plan	3/3	3/3	5.0	4.7
cat.OE4.plan	3/3	—	5.0	—
cat.OE5.plan	0/3	—	—	—
cat.OE6.plan	0/3	—	—	—
cat.ST1.plan	0/3	3/3	2.3	4.3
cat.ST2.plan	3/3	3/3	4.3	5.0
cat.ST3.plan	1/3	1/3	3.3	3.3
lp.CTO.agent	0/3	3/3	4.7	4.0
lp.CTO.plan	0/3	3/3	4.0	4.7
lp.LP1b.agent	0/3	3/3	4.0	4.3
lp.LP1b.plan	0/3	3/3	4.0	4.7
lp.LPR1.agent	0/3	3/3	3.0	4.3
lp.LPR1.plan	0/3	2/3	2.7	3.7
r5.122633.LP	0/3	3/3	4.7	5.0
r5.122633.LV7to5	3/3	—	4.0	—
r5.122633.MAIN	3/3	—	5.0	—
r5.122633.QG	3/3	—	5.0	—
r5.122633.VOCAB	3/3	3/3	4.7	5.0
r5.122633.WHY	3/3	—	5.0	—
r5.29060.LP	0/3	—	4.0	—
r5.29060.LV4to2	3/3	—	4.0	—
r5.29060.LV4to6	3/3	—	4.7	—
r5.29060.MAIN	3/3	3/3	5.0	5.0
r5.29060.QG	3/3	—	4.7	—
r5.29060.VOCAB	3/3	—	4.3	—
r5.29060.WHY	3/3	3/3	5.0	5.0
r5.29189.LP	1/3	—	4.7	—
r5.29189.LV3to1	3/3	—	5.0	—
r5.29189.MAIN	3/3	—	5.0	—
r5.29189.QG	3/3	3/3	4.7	5.0
r5.29189.VOCAB	3/3	—	5.0	—
r5.29189.WHY	3/3	—	5.0	—
r5.41150.LP	1/3	—	4.7	—
r5.41150.LV1to3	3/3	—	4.0	—
r5.41150.MAIN	3/3	—	5.0	—
r5.41150.QG	3/3	—	5.0	—
r5.41150.VOCAB	3/3	3/3	4.3	5.0
r5.41150.WHY	3/3	—	5.0	—
r5.45741.LP	0/3	—	4.0	—

Task	before well	after well	before judge	after judge
r5.45741.LV5to3	3/3	—	4.0	—
r5.45741.MAIN	3/3	—	4.3	—
r5.45741.QG	3/3	—	4.7	—
r5.45741.VOCAB	3/3	—	4.7	—
r5.45741.WHY	2/3	3/3	3.7	4.7
r5.46270.LP	0/3	3/3	4.3	4.3
r5.46270.LV6to4	3/3	—	4.0	—
r5.46270.LV6to8	3/3	—	5.0	—
r5.46270.MAIN	3/3	—	5.0	—
r5.46270.QG	3/3	—	4.3	—
r5.46270.VOCAB	3/3	—	4.7	—
r5.46270.WHY	3/3	—	4.0	—
r5.92658.LP	1/3	—	4.3	—
r5.92658.LV2to4	2/3	—	3.7	—
r5.92658.MAIN	3/3	3/3	4.7	5.0
r5.92658.QG	3/3	—	5.0	—
r5.92658.VOCAB	3/3	—	4.3	—
r5.92658.WHY	3/3	—	4.7	—
r5.94818.LP	0/3	—	4.7	—
r5.94818.LV8to10	3/3	—	4.7	—
r5.94818.LV8to6	3/3	—	4.7	—
r5.94818.MAIN	3/3	—	4.3	—
r5.94818.QG	3/3	3/3	4.7	5.0
r5.94818.VOCAB	3/3	—	5.0	—
r5.94818.WHY	3/3	—	5.0	—
ra.RA02.agent	—	3/3	—	5.0
ra.RA03.agent	—	3/3	—	5.0
ra.RA04.agent	—	3/3	—	5.0
ra.RA05.agent	—	3/3	—	5.0
ra.RA07.agent	—	3/3	—	5.0
ra.RA09.agent	—	3/3	—	5.0
ra.RA11.agent	—	2/3	—	5.0
ra.RA13.agent	—	2/3	—	4.3
ra.RA14.agent	—	3/3	—	4.7
ra.RA15.agent	—	3/3	—	4.7
ra.RA16.agent	—	3/3	—	4.7
ra.RA17.agent	—	3/3	—	5.0
ra.RA18.agent	—	3/3	—	5.0
ra.RA20.agent	—	3/3	—	5.0
ra.RA21.agent	—	2/3	—	4.0
ra.RA23.agent	—	3/3	—	5.0
ra.RA24.agent	—	3/3	—	5.0
ra.RA25.agent	—	3/3	—	5.0
ra.RA26.agent	—	3/3	—	5.0
ra.RA27.agent	—	3/3	—	4.7
ra.RA28.agent	—	3/3	—	5.0
ra.RA29.agent	—	3/3	—	4.7
ra.RA30.agent	—	3/3	—	4.7
ra.RA31.agent	—	3/3	—	5.0
ra.RA32.agent	—	3/3	—	5.0
ra.RA33.agent	—	3/3	—	5.0
ra.RA35.agent	—	3/3	—	5.0
ra.RA36.agent	—	3/3	—	5.0
ra.RA37.agent	—	3/3	—	5.0
ra.RA38.agent	—	2/3	—	4.3
ra.RA39.agent	—	3/3	—	5.0
ra.RA40.agent	—	3/3	—	5.0
ra.RA41.agent	—	3/3	—	5.0
ra.RA42.agent	—	3/3	—	5.0
rel.A01	10/10	3/3	4.8	5.0
rel.A02	2/10	—	4.4	—
rel.A03	10/10	3/3	5.0	5.0
rel.A04	10/10	—	—	—
rel.A05	10/10	3/3	—	—
rel.A06	10/10	—	—	—
rel.C01	10/10	3/3	—	—
rel.G01	10/10	3/3	5.0	5.0
rel.G02	10/10	3/3	5.0	5.0
rel.G03	10/10	—	—	—
rel.N01	10/10	3/3	5.0	5.0
rel.P01	10/10	3/3	5.0	5.0
rel.P02	10/10	3/3	—	—
rel.S01	0/10	3/3	2.9	4.7
rel.S02	10/10	3/3	—	—
rel.T01	10/10	3/3	—	—
rel.T02	10/10	3/3	5.0	5.0
rel.X01	10/10	3/3	—	—

Turn outcomes (v2)

Family	complete	paused	stopped	error	idle_timeout	tool_error	no_turn_end
catalogue	87	3	0	0	0	0	0
lesson_plan	18	0	0	0	0	0	0
r5	30	0	0	0	0	0	0
real_asks	68	34	0	0	0	0	0
reliability	36	12	0	0	0	0	0

First content (v2 telemetry `first_content_ms`; not a gate)

Turns with a value: 245; turns whose telemetry row lacks the key: 0.

Measure	p50	p95	n
first_content_ms (telemetry, all)	3.2s	7.1s	245
first_content_ms (cold / warm p50)	3.3s / 3.2s	—	42 / 203
ttft_any_ms (eval, client-side, same turns)	3.2s	7.2s	245

Engine stage timings (v2, from artifact.end, seconds from tool start; not a gate)

Engine	n	resources p50/p95	engine_start p50/p95	first_token p50/p95	first_item p50/p95	draft p50/p95	write p50/p95	check p50/p95	verify p50/p95	total p50/p95	cache_read (median sum)
generate_questions	12	—	0.0/0.0	1.9/2.2	3.4/3.7	—	7.6/8.9	—	2.2/3.0	12.2/16.5	0.0
write_artifact (lesson_plan)	18	0.0/0.0	0.0/0.0	2.0/3.0	—	19.2/24.1	—	1.1/3.8	—	21.8/26.5	1098.0
write_artifact (weekly_plan)	6	0.0/0.3	0.0/0.3	1.9/2.2	—	25.6/29.2	—	0.9/5.4	—	28.9/32.3	1716.0

Context degradation (v2, silent to the teacher)

Telemetry rows joined: 288/288 v2 turns (missing = no tokenz row for that turn_id).

Rate	Measured	How
Resource lookup failed	0/288 (0.0%)	`ctx_resource_lookup_failed` truthy, over turns that report the field
History summary failed	0/288 (0.0%)	`ctx_summary` equals failed, over turns that report the field
Article truncated (notice shown)	0/117 (0.0%)	`ctx_article_notices > 0`, over turns with article text

Tool-name corrections (v2, the model called a capability id as a tool name)

Key `tool_name_corrections` (names `tool_names_corrected`): 0/288 turns with ≥1 correction (0.0%); 0 corrections in total. Unjoined turns (no telemetry row): 0. **Contract breaches (row present, key missing or non-integer): 0** ✅.

Family	Turns	With correction	Corrections
catalogue	90	0 (0%)	0
lesson_plan	18	0 (0%)	0
r5	30	0 (0%)	0
real_asks	102	0 (0%)	0
reliability	48	0 (0%)	0

Outcome	Turns	With correction
complete	239	0 (0%)
paused	49	0 (0%)

LP8: old vs corrected criterion

Corrected 2026-09-15 (coordinator-approved): a card with no current week cannot resolve "plan next week", so asking which week with a pending_prompt whose options name a real week is correct; a guessed-week plan, a hedge or a prose question fails.

On an article card that names the next week (nextWeekId), the pass is a weekly plan for that week naming at least 2 of its real titles; asking with real options there is the soft column, not a pass.

Task	Turns	Old criterion pass	Corrected pass	Asked, real options	Asked, no real option	Planned a guessed week	Planned the named next week	Asked, real options (soft)	Hedge / other
cat.LP8.plan	3	1	2	2	0	1	0	0	0

Per task

Task	Ask	n	Det.	Well	Judge	Behaviours	Common failures	Example (first failing run)
cat.APP1.plan	how do I assign this article to my class?	3	3	3	4.3	1		
cat.APP4.plan	how do I create a new classroom?	3	3	2	3.7	1		complete: To create a new classroom, use the "Start creating a new classroom" action, which opens th
cat.APP5.plan	how do I add a student to my class?	3	3	3	4.3	1		
cat.AS2.plan	make a quiz on this article	3	3	3	4.7	1		
cat.AS4.plan	make question 3 easier	3	3	3	4.3	1		
cat.AS8.plan	write feedback comments for the students who sco	3	3	3	4.7	2		
cat.AS9.plan	create an assessment covering week 1	3	1	0	2.3	1	questions_present×2	complete: Creating a week 1 assessment on *Our Nation Studies Weekly – American Foundations.* Week
cat.CD1.plan	who is falling behind?	3	3	3	4.7	1		
cat.CD6.plan	based on these results, what should I reteach?	3	3	3	5.0	1		
cat.CD7.plan	how is Student 1 doing?	3	3	2	3.7	1		complete: Here's a picture of Student 1's progress in *Our Nation Studies Weekly*, based on the repo
cat.CD8.plan	what do I need to grade?	3	3	3	4.3	1		
cat.CM2.plan	create a vocabulary activity from this article	3	3	3	4.0	1		
cat.CM4.plan	make a graphic organizer for this article	3	3	3	5.0	1		
cat.CM5.plan	create a rubric for a paragraph response about t	3	3	3	4.3	1		

Task	Ask	n	Det.	Well	Judge	Behaviours	Common failures	Example (first failing run)
cat.CM9.plan	make a vocabulary list with kid-friendly definit	3	3	3	4.7	1		
cat.CO1.plan	write an email to parents about what we're learn	3	3	3	4.7	1		
cat.CO2.plan	draft a note to Student 8 reminding them to turn	3	3	3	4.0	1		
cat.CO4.plan	write a short newsletter paragraph about what ou	3	3	3	4.7	1		
cat.LP1.agent	create a lesson plan for the article	3	3	3	4.0	1		
cat.LP1.plan	create a lesson plan for the article	3	3	3	4.0	1		
cat.LP3.agent	plan a lesson using this	3	3	3	4.0	1		
cat.LP3.plan	plan a lesson using this	3	3	2	4.0	1		complete: I'll write a lesson plan built from this article on producers, consumers, and shortages.
cat.LP5.plan	How should I differentiate this article for my s	3	3	3	5.0	1		
cat.LP8.plan	Help me plan next week from this publication.	3	2	1	3.3	3	asks_which_week_with_real_options×1	complete: I'm creating a week-long lesson plan based on *Our Nation Studies Weekly – American Founda
cat.OE1.plan	what is a shortage?	3	3	3	5.0	1		
cat.OE2.plan	why do prices go up when there is a shortage?	3	3	3	4.7	1		
cat.OE3.plan	give me a real-world example of producers and co	3	3	3	4.7	1		
cat.ST1.plan	what standards does this article cover?	3	3	3	4.3	2		
cat.ST2.plan	what standards are covered this week?	3	3	3	5.0	1		
cat.ST3.plan	align this article to Common Core standards	3	3	1	3.3	2		complete: Here's an alignment for "Producers and Consumers" to well-known Common Core codes for grad
lp.CTO.agent	Create a lesson plan for this article.	3	3	3	4.0	1		
lp.CTO.plan	Create a lesson plan for this article.	3	3	3	4.7	1		
lp.LP1b.agent	create a lesson plan for this article	3	3	3	4.3	1		
lp.LP1b.plan	create a lesson plan for this article	3	3	3	4.7	1		
lp.LPR1.agent	create a lesson plan for this week	3	3	3	4.3	2		
lp.LPR1.plan	create a lesson plan for this week	3	3	2	3.7	2		complete: I'll build this from the Week 3 "Water on Earth" resources on screen — the lesson plan pan
r5.122633.LP	Create a lesson plan for this article.	3	3	3	5.0	1		
r5.122633.VOCAB	What does "empirical" mean in this article? Expl	3	3	3	5.0	1		
r5.29060.MAIN	What is the main idea of this article? Give me a	3	3	3	5.0	1		
r5.29060.WHY	Why are fossils helpful to paleontologists?	3	3	3	5.0	1		
r5.29189.QG	Create 5 questions for my class from this articl	3	3	3	5.0	1		
r5.41150.VOCAB	What does "tilt" mean in this article? Explain i	3	3	3	5.0	1		
r5.45741.WHY	Why were the Spanish and French fleets fighting	3	3	3	4.7	1		
r5.46270.LP	Create a lesson plan for this article.	3	3	3	4.3	1		
r5.92658.MAIN	What is the main idea of this article? Give me a	3	3	3	5.0	1		
r5.94818.QG	Create 5 questions for my class from this articl	3	3	3	5.0	1		
ra.RA02.agent	Show inactive classrooms	3	3	3	5.0	1		
ra.RA03.agent	search this page for Demo	3	3	3	5.0	1		
ra.RA04.agent	Show active classrooms	3	3	3	5.0	1		
ra.RA05.agent	can you fill it for me	3	3	3	5.0	2		
ra.RA07.agent	Are there any ties in the leaderboard, and who i	3	3	3	5.0	1		
ra.RA09.agent	is anything else still needed before I can see t	3	3	3	5.0	1		
ra.RA11.agent	what is in the Training category?	3	2	2	5.0	1	no_refusal×1	paused, action training.openCategory: I can't see what's inside the Training category from here — the page only shows category n
ra.RA13.agent	How many separate pieces of customized content h	3	2	2	4.3	1	no_refusal×1	complete: This report shows standards performance, not a list of assigned customized content items —
ra.RA14.agent	What are my three most recent conversations with	3	3	3	4.7	1		
ra.RA15.agent	Has anybody left this class? If so tell me who,	3	3	3	4.7	1		
ra.RA16.agent	Which of my active classrooms has the most publi	3	3	3	4.7	1		
ra.RA17.agent	Walk me through what happens on the fifth screen	3	3	3	5.0	1		
ra.RA18.agent	what is this draft called?	3	3	3	5.0	1		
ra.RA20.agent	what is in unit 2?	3	3	3	5.0	1		
ra.RA21.agent	Rank this pupil's weeks from strongest to weakes	3	3	2	4.0	1		complete: Based on assessment scores, activity performance, and engagement across all the tracked me
ra.RA23.agent	according to this page, why is the profile form	3	3	3	5.0	1		
ra.RA24.agent	Put the whole list in order by one property Grad	3	3	3	5.0	2		
ra.RA25.agent	Which fields on this page can I edit and which a	3	3	3	5.0	1		
ra.RA26.agent	create and assign this customized content now	3	3	3	5.0	1		
ra.RA27.agent	How much of the publication has the class finish	3	3	3	4.7	1		
ra.RA28.agent	list my highlights	3	3	3	5.0	1		
ra.RA29.agent	If I answer a quiz question while I am impersona	3	3	3	4.7	1		
ra.RA30.agent	How many things are in that list altogether?	3	3	3	4.7	1		
ra.RA31.agent	which assessment went worst?	3	3	3	5.0	1		
ra.RA32.agent	how many students have handed this in?	3	3	3	5.0	1		

Task	Ask	n	Det.	Well	Judge	Behaviours	Common failures	Example (first failing run)
ra.RA33.agent	pause the video	3	3	3	5.0	1		
ra.RA35.agent	Which kinds of content is this student answering	3	3	3	5.0	1		
ra.RA36.agent	open the previous school year notes	3	3	3	5.0	1		
ra.RA37.agent	Tell me about clouds in general, weather pattern	3	3	3	5.0	1		
ra.RA38.agent	How many students have completed this, how many	3	3	2	4.3	1		complete: 3 of 8 students have completed (turned in) the assessment, 1 is in progress, and 1 has not
ra.RA39.agent	switch to my other classroom	3	3	3	5.0	1		
ra.RA40.agent	open the first rubric you can find on this page	3	3	3	5.0	1		
ra.RA41.agent	list the teacher resource panels on this page	3	3	3	5.0	1		
ra.RA42.agent	hide week 1 assessments from students	3	3	3	5.0	1		
rel.A01	What is this article about?	3	3	3	5.0	1		
rel.A03	Create 3 multiple choice questions from this art	3	3	3	5.0	1		
rel.A05	Find me articles about the water cycle.	3	3	3	—	1		
rel.C01	Delete dsfsdf.	3	3	3	—	1		
rel.G01	What do I have waiting to be graded?	3	3	3	5.0	1		
rel.G02	What is the class average this week?	3	3	3	5.0	1		
rel.N01	how many submissions are waiting to be graded in	3	3	3	5.0	1		
rel.P01	how many students have a verified family email?	3	3	3	5.0	1		
rel.P02	drop Student 4 from this class	3	3	3	—	1		
rel.S01	How do I add a co-teacher to my classroom?	3	3	3	4.7	1		
rel.S02	Change the grade to Fifth Grade.	3	3	3	—	1		
rel.T01	take me to week 5	3	3	3	—	2		
rel.T02	what is hidden from students right now?	3	3	3	5.0	1		
rel.X01	Find me articles about the branches of governmen	3	3	3	—	2		

Scout prompt architecture: review and recommendation (2026-10-03)

Design only; no product code or model calls, $0. Code: rag-merge `a0d80c4c`, swo `0706749415`. Real calls: `prompt-anatomy-2026-10-03.md` and its examples (dev log, includes evaluation batches). Tokens are characters ÷ 3.13, the ratio measured on the chat prefix (55,409 chars = 17,685 tokens); treat every converted figure as **±15% until a CountTokens run settles it**. Prices are Bedrock-billed Sonnet 5 rates (`core/config.py:32`): $2.20 in, $11 out, $0.22 cache read, $2.75 cache write per MTok.

Summary for Salman

- **The chat loop is the problem; the engines are not.** Every chat call carries the same 17.7k tokens (8.8k of rules, 8.6k of tool definitions) before the page. Engines send 0.2k–2k tokens of rules each. The chat call was 66% of the questions chain's cost.
- **Most of the 17.7k is "just in case".** A tab switch needs about 1.2k of the 8.8k rule tokens and 1 of 18 tools.
- **Caching already hides much of the size on warm turns.** A warm 21.5k prompt cost $0.0062. Size costs most on a cold cache (9.5× more), and in attention: one long rulebook makes rules compete.
- **A router call (your idea) does not pay on Bedrock with caching.** It adds a call (about $0.002, 1–2 s) to every turn to save about $0.002 of cache reads on a warm turn. It only wins on cold turns. Scout's deterministic `direct_dispatch` is already the cheap version of it.
- **Recommendation: "rules travel with the feature."** A small shared core of about 2.5k tokens goes in all five emitters (chat and every engine). Each feature's rules move to where they appear only when the feature is in play: its tool description or result, its page capability, or the data block it governs. A `read_guide` tool covers rare topics. The prefix stops growing with features.
- **Expected change per turn:** input cost –35% on a warm call, –45% on a cold start, and about the same on the first call on a new page, which the page card dominates (G-CARD is measuring that lever now). Estimates, not results.

1. Inventory

1.1 The chat loop (`core/runtime.py:_system_blocks`, `core/loop.py:build_request`)

Cache points today: after tools, after the constitution, after the page layer, and on the last history message: 4 of 4 allowed (`loop.py:179`). Teacher, scope, summary and mode blocks are uncached.

Layer	Size (switch tab / questions example)	Cached	Same for every teacher?
Tools, 18	27,742 chars, about 8.6k tok	yes, own point	yes
Constitution	27,667 chars, about 8.8k tok	yes, own point	yes
Page card	4,766 / 7,481 chars	yes, per page	no
Actions list	5,710 / 14,581 chars	yes, per page	no
Article	0 / 1,892 chars	yes, per page	no
"About you" teacher block	not present / 1,811 chars	**no, re-sent fresh every call**	per teacher, stable
History summary, mode, scope, page search	0.8k–1.9k chars	no	no

Constitution by topic (characters, about tokens): facts and sources (scope rule and its decline replies, craft, difficult history) 8,166 (2.6k); making things 4,775 (1.5k); answering questions 4,437 (1.4k); trust and privacy 2,784 (0.9k); acting in the app 2,560 (0.8k); finding things 2,412 (0.8k); voice and format 1,255 (0.4k); grading 611 (0.2k); who you serve and intro 773 (0.25k).

Tools by size (compact JSON chars): `draft_assignment` 3,456, `generate_questions` 3,291, `write_artifact` 2,356, `search_curriculum` 1,783, `level_article` 1,739; the other 13 are 517–1,575 each. Descriptions are 13.9k of the 27.0k; the rest is schemas.

Needed per turn vs just in case (my reading of the text against three asks; a judgement, not a measurement):

Ask	Rule tokens it needs	Tools it needs
Switch tab	acting in the app, trust, voice: about 1.2k of 8.8k (14%)	`page_action` (290 of 8.6k)
"Make 5 quiz questions"	making, sources, voice, trust: about 3k (35%)	`generate_questions`
Question about the article	sources, answering, voice, trust: about 5.3k (60%)	none, or `get_article_resources`

Repeated text. The "Use the specialised tools" paragraph (2,255 chars) restates what the tool descriptions say. `SCOPE_RULE` (1,561) is in the constitution and in two engines. The teacher block is stable per teacher but is billed at full input price on every call.

Cache behaviour seen in logs. A mid-turn Haiku step cannot read Sonnet's cache and wrote its own 21,333 tokens ($0.029, 10-02 10:31:42). Cache minimums are 1,024 tokens on Sonnet 5 and 4,096 on Haiku 4.5 (claude-api skill, `shared/prompt-caching.md`), so the leveller (600 tokens) and every Haiku checker never cache. In the dev window, cache reads were 46% of chat spend, cache writes 40% and output 13% (1,458 dev Sonnet chat calls, $15.26; 17 bodiless calls excluded).

1.2 The engines (each builds its own system prompt; no constitution)

Emitter	Model	System chars (tok)	Rest of input	Cache
compose / `write_artifact` (lesson plan, worksheet, …)	Sonnet	`SYSTEM` 6,071 (1.9k), incl. `SCOPE_RULE`; plus one `KIND_SPECS` entry, 103–3,422 chars	article, resources, brief	system point
questions	Sonnet	3,064 (1.0k), half of it `SCOPE_RULE`	article	write 1,266, read on repair
questions verifier	Haiku	996 + article	items	below minimum
level	Sonnet	644 (0.2k); check 1,011 Haiku	article	none (below minimum)
grade, written	Sonnet	1,670 (0.5k)	answers	none
grade, drawings	Sonnet	2,396 (0.8k)	answers + image (about 1.5k tok)	none
group, misconceptions	Haiku	1,314 / 1,226	ideas	none
standards	Sonnet	1,949 + 593	items	none
prepare_lesson write; lead-in sentence	Sonnet; Haiku	1,597; about 600	resources; ask	none

Compose already uses the pattern recommended below: the kind's spec is loaded only for that kind (`prompts/engines/compose.py:140`). The engines are lean; what they lack is a **shared core**. Trust, voice, student-wording and scope rules reach some of the five emitters and not others.

2. Options

Per-call costs, with a page layer of 4k tokens and 150 output tokens:

Scenario	Today	(a+e) core 6.5k	(c) router + 8k main
Warm (follow-up, same page)	$0.0064	$0.0042	$0.0062
Shared layer warm, new page	$0.0166	$0.0168*	$0.0200
Cold start	$0.0613	$0.0330	$0.0370
Added latency	—	none	+1–2 s every turn

*Modules keyed on the page land in the page layer, so a new-page call stays about the same. Only a smaller page card moves this row.

(a) Modular constitution. A core of about 2.5k tokens (who, trust and privacy, voice, the three-source rule, never claim an unverified action). Topic modules are loaded **deterministically** from the page card, mode or data present: student data brings the pseudonym and describe-not-label module, an article brings article answering, `search.find` brings the page-search line. This is ONLINE-7581's mechanism, which passed its Search-page lines (17/18 vs 2/18). Ask-dependent topics come **on demand** through a `read_guide(topic)` client tool, listed in the core as one line per topic. Works on Converse (an ordinary client tool). Risk: the model skips `read_guide` and silently answers without the rules, so measure per topic. Build: about 2 sessions. A new feature adds a module, not prefix.

(b) Deferred tools or tool search. Not available to Scout as built: tool search on Bedrock is InvokeModel only, not Converse (skill `shared/platform-availability.md`), and Scout calls ConverseStream. With 18 tools it would also add a round-trip for little gain, and swapping the tool list per turn breaks the tools cache. The workable equivalent is **shorter tool descriptions**: the "when to call" stays in the description, the "what to say after" moves into the tool result. Target about 4k tokens, from 8.6k.

(c) Two-stage router. Priced above. The router's prompt (about 1.5k tokens) is below Haiku's cache minimum, so it is paid fresh every turn, and it adds 1–2 s (dev Haiku checker calls took 2.1 s). It saves tokens that caching already discounts by 90%, and a misroute removes rules the main model needed. Scout already has the cheap deterministic version (`direct_dispatch`, 0 misroutes over 112 eval and 58 real asks, `runtime.py:1931`). **Not recommended.**

(d) Unify engines with the chat loop. Engines should **not** take the chat prompt: that would add about 9k tokens to every engine call for rules about tools and page actions they never use. The reverse is right: **one shared core module included by all five emitters**, then each emitter's own task layer.

(e) Trimming. Remove the restated tool guidance, merge the scope procedure into the core, and cut the rules that exist "just in case". Check each cut against `MEASURED-AND-FAILED.md`: several sentences were earned by gates. On its own, trimming shrinks the prompt once but does not stop it growing.

Cache layout fixes (rule text unchanged). - (i) Move the teacher block into a cached per-teacher layer. - (ii) Use one breakpoint for tools plus core, since tools render first. That frees a point for the per-teacher layer. - (iii) Stop the Haiku mid-turn step from writing a full Sonnet-sized cache. - (iv) Give the shared core a 1-hour TTL if Converse accepts it. Bedrock supports 5 m and 1 h caching per the skill, but I did not confirm the Converse `cachePoint` TTL field; one call settles it. At internal-only traffic the 5-minute shared layer is likely cold often. Not measured; ONLINE-7526's cache fingerprints in prod logs can settle it.

Recommended combination: (e) + (a, deterministic first, then on demand) + (b as shorter descriptions) + (d as a shared core) + the cache fixes. Not (c).

3. Recommended architecture

Layers, in cache order:

1. **Tools**, about 4k tok. Each description says when to call the tool, in about 300 characters at most. Fixed order, shared by all teachers.
2. **Scout core**, about 2.5k tok, the same text in all five emitters. The chat loop adds an addendum of about 0.3k: the topic index for `read_guide`, Agent and Plan modes, and the never-claim-done rule. Breakpoint 1 sits here (tools plus core), on the 1-hour TTL if available.
3. **Teacher layer:** the "About you" and memory blocks. Stable per teacher. Breakpoint 2.
4. **Page layer:** page card, article, capability lines, and the **deterministic modules the page triggers**. A capability line carries its id, a short description and its arguments; its routing rules come back in `page_action`'s result. Breakpoint 3.
5. **Live:** mode, scope, history summary.
6. **History:** includes any `read_guide` results and tool results carrying "how to report this" text. Breakpoint 4.

How a feature adds its piece: it adds a capability line (page layer, only on pages that offer it), a tool or a tool-result note (only when it runs), a module keyed on the page or data that needs it, or a `read_guide` topic of one index line. Only a rule that must hold on every turn of every emitter enters the core; a pin holds core plus tools ≤ 7k tokens, each module ≤ 1k. So features are unlimited, and each costs tokens only on the turns that use it.

History replay: deterministic modules are rebuilt from the stored page state, as the card is today. `read_guide` output is an ordinary stored tool result, so a thread replays identically. Store the guide's hash with it.

Guardrail: system-side modules do not change what the guardrail sees. **One unverified risk:** `read_guide` results arrive as tool results in messages. Whether the input guardrail scans tool results on Converse (and bills them) is not settled here; module text about difficult history could trip a topic policy. Check with `ApplyGuardrail` on the guide texts (cents) before building the on-demand path. The deterministic path has no such risk.

4. Measurement plan (declared for later; not run)

- **Population, frozen before any arm row:** 240 first-turn asks. 80 action (G-CARD's P/U/N/Z strata), 40 making, 40 article and help, 30 scope (incl. difficult history), 30 student-data; plus 40 two-turn threads. Each ask's accept set and expected modules are written before any row.
- **Arms:** rag arm sha against control sha, both named at freeze (never `HEAD`), served from one snapshot port (not `:8002`); swo sha identical on both sides. Every `CORE_*` flag and the guardrail version set explicitly on both sides. Rows grouped by page, sides alternating, as G-CARD does; plus 20 declared cold asks (after more than 5 min idle).
- **Deciding lines:**
- **L1 task success:** deterministic, paired, all 240: the first tool or `page_action` id is in the accept set, and a scope ask ends in the decline reply. Pass if control-only minus arm-only ≤ 6.
- **L2 routing to rules:** per topic, the guide or module is loaded when its label says it is needed. Reported, and deciding at ≥ 90% for scope and student-data topics.
- **L3 tokens:** median input-side tokens per first turn (fresh + read + write) at least 35% lower. Deterministic; computable offline with CountTokens before any model call.
- **L4 cost:** per completed turn, warm and cold reported separately; the arm is no higher on either.
- **L5 latency:** p50 ≤ +0.3 s and p95 ≤ +1.0 s against control.
- **Do-no-harm screens (not deciding alone):** `name_claim`, `student_wording`, `leak`, `ability_claim`, `standards_claim`, the questions checker's reject rate, and a Sonnet side-hidden judge on making and answer replies. Arm HARM ≤ control + 2. Each screen first proved able to fire on a known positive.
- **Power:** assuming 12% discordant pairs (G-CARD's guess), n=240 catches a true 4-point drop about 80% of the time and an 8-point drop about 99% (scaled from G-CARD's exact figures at n=160: 66.7% and 97.8%). Recompute exactly before freezing. p95 detects shifts of about 1 s, not less.
- **Cost:** about 560 chat calls (about $0.012 warm, $0.035 cold) is about $9. About 160 engine chains at about $0.03 is $5. Judge about $2. **About $16 total, over the $10 line, so it needs coordinator approval.** The offline token pass is $0.

5. Phasing and size

Phase	What	Risk	Sessions
0	CountTokens inventory; prompt audit against `MEASURED-AND-FAILED.md`; `ApplyGuardrail` on guide texts; Converse TTL test	none	1 Sonnet
1	Cache fixes (i)–(iv): rule text unchanged, only block order and cache points; verify by cache fingerprints and a text-identity pin	low	1 Sonnet
2	Shared core, deterministic modules, short tool descriptions, trim; §4 gate	medium: every turn and emitter	1 Opus design, 2 Sonnet build, 1 Sonnet measure
3	`read_guide`, only if Phase 2 leaves just-in-case rules	medium	1–2

Before go-live: Phase 0 is safe and Phase 1 is low risk. Phase 2 is a behaviour change on every turn, so it either lands under the gate before the QA handoff resumes or waits until after go-live. It must not land in between untested. Risks: gate-earned sentences lose effect when moved; the new-page row needs a smaller action card (G-CARD); edits under `lambdas/ai_assistant` reload `:8002`.

G-CARD: running at 10:40 PKT (116 of 160 rows per side); outcomes not read.

6. Questions for Salman

1. **Direction?** (A) Rules travel with the feature, plus cache fixes **[recommended]**. (B) Two-stage router. (C) Trim only. (D) No change now.
2. **When?** (A) Phases 0–1 now; Phase 2 built and gated now, merged only on a pass; go-live does not wait **[recommended]**. (B) Everything after go-live. (C) Phases 0–1 only.
3. **Engines?** (A) One shared core module in all five emitters **[recommended]**. (B) Leave engines as they are.
4. **Gate spend?** (A) Approve up to $20 for the §4 gate **[recommended]**. (B) Approve the $0 offline token pass only, decide later.
5. **Ticket?** (A) A new Jira ticket for the prompt restructure, with Phase 1 as its first commit **[recommended]**. (B) Fold it into ONLINE-5385.